

GhostRes Edge

Local AI video restoration with a
verified Snapdragon model path

Short Pitch Presentation

Kartikey Bhardwaj | B.Tech CSE, DIT University

Low-resolution footage needs more than pixel stretching

Traditional interpolation increases size, but it loses edge definition and texture.

Video quality

480p and compressed recordings become soft or blocky when enlarged.

Privacy

Cloud workflows can require creators to upload personal footage before enhancement.

Control

Creators need a visible speed-versus-quality choice instead of an opaque process.

GhostRes Edge uses Real-ESRGAN super-resolution locally and reports the resulting processing tradeoffs.

A local desktop workflow that exposes the tradeoff

The user selects a video, chooses an enhancement mode, runs restoration, and reviews measured output.

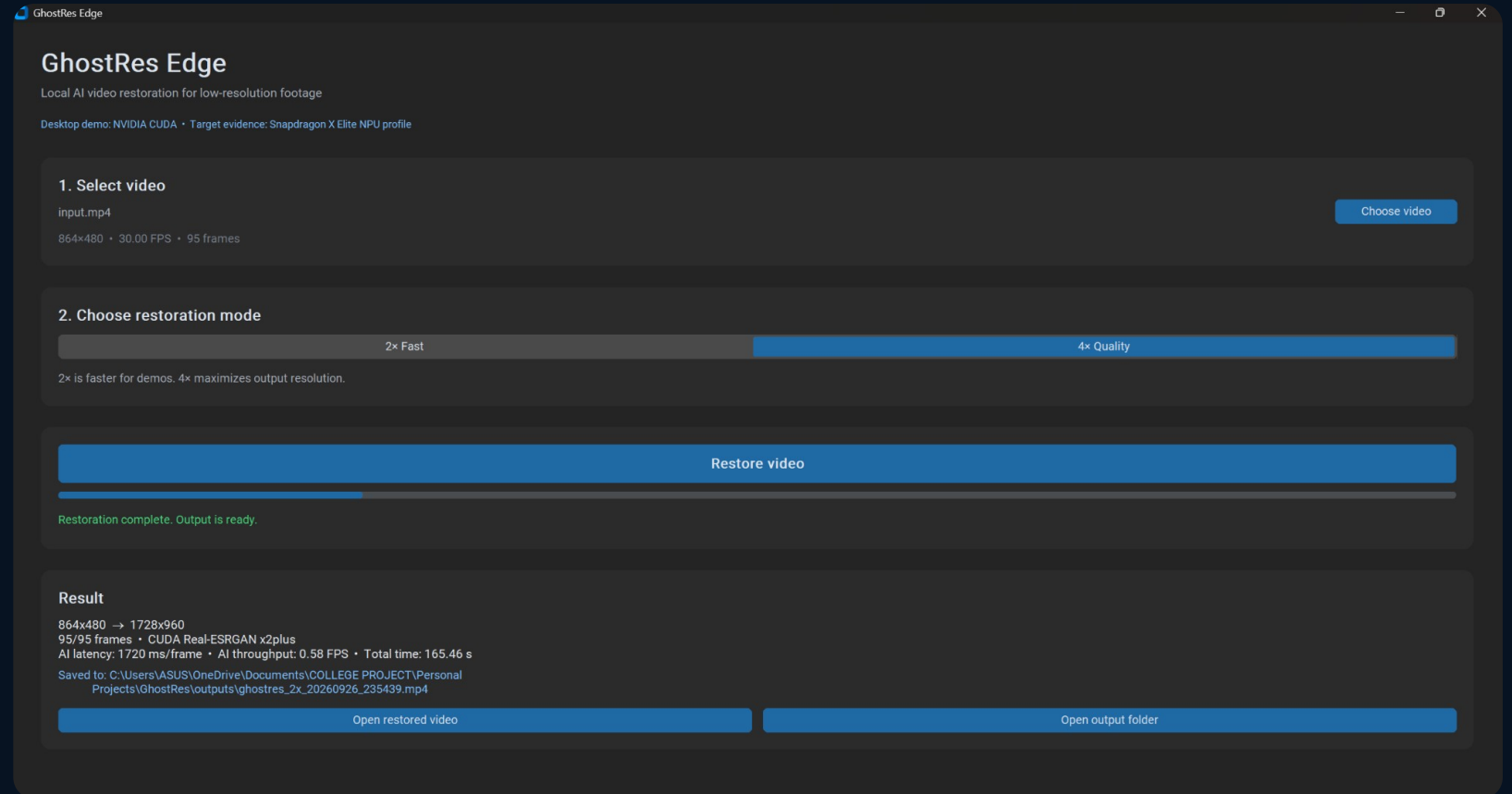
Fast 2x

Lower output resolution with quicker restoration.

Quality 4x

Higher output resolution with greater compute cost.

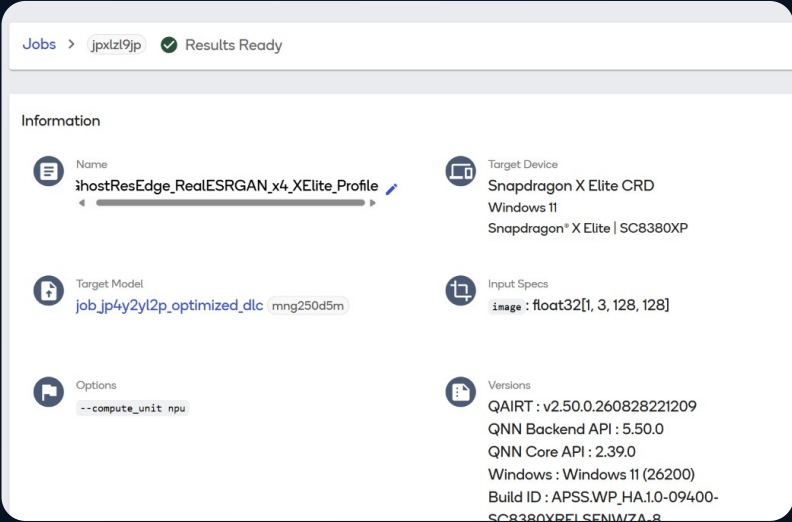
The UI reports output resolution, frames processed, AI latency, AI FPS, and total time.



Desktop video result and Snapdragon model profile

The two measurements answer different questions and use different workloads.

Environment	Workload	Measured result
RTX 3060 desktop app	95 frames at 864x480 Fast 2x	1728x960 output 165.46 seconds 1720 ms per frame 0.58 AI FPS
Snapdragon X Elite CRD	Real-ESRGAN x4 ONNX tile float32 [1, 3, 128, 128] QNN NPU	65.4 ms inference 0.2 MB peak memory



Qualcomm AI Hub profile result summary

The desktop result measures an entire video. The Snapdragon result measures one 128x128 x4 model tile.

Modular inference architecture

A reusable interface keeps model code separate from the video workflow.

Desktop workflow

CustomTkinter UI
OpenCV video processing
Resolution checks and
performance telemetry

EnhancementEngine

Single interface for frame enhancement

Current engine

CudaRealESRGANEngine
x2plus and x4plus models
128 pixel tiled inference

Snapdragon path

ONNX export
QNN NPU profile on X Elite
CRD
Future OnnxQnnEngine

The future QNN backend can retain the existing UI and video pipeline rather than requiring a product rewrite.

What GhostRes Edge demonstrates today

A working local restoration application with measured video output and verified Snapdragon model profiling.

Available now

Working desktop app
Fast 2x and Quality 4x modes
Full 95-frame local restoration result
ONNX model validated with
onnx.checker
Snapdragon X Elite CRD QNN profile

Next engineering milestone

Implement the QNN inference
engine inside the same application

Benchmark the full video pipeline on
a physical Snapdragon PC

Preserve source audio during output
creation

GhostRes Edge

Local AI video restoration built with a clear route from CUDA development to Snapdragon QNN deployment.